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Dielectric Properties of LiNbO_3 Single Crystal up to 9 Gc

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Lithium metaniobate LiNbO_3 is a ferroelectric material belonging to crystal point group C_{3v} and recently many interests have been paid to its properties from the fundamental and practical viewpoints. This note presents the experimental results on the frequency and temperature dependences and anisotropy of its dielectric properties.

Examined specimens were x - and z - plates up to 900 Mc, and x - and z - bars for the microwave region. Admittance bridge was used in the frequency range from 10 kc to 30 Mc, Q meter from 20 to 70 Mc and Thurston type comparator from 100 to 900 Mc. Perturbation method served in the measurements of 4 Gc and 9 Gc.

The change in dielectric constant ϵ_{11} and ϵ_{33} with frequency is shown in Fig. 1. In the frequency range below 1 Mc, these two keep constant values corresponding to the mechanically free state, that is, ϵ^T . When the frequency reaches one of acoustic resonant modes, the dielectric constant steeply increases and thereafter abruptly falls down to minimum where the frequency corresponds to antiresonance. From these frequency series of resonance and antiresonance, acoustic phase velocities and electromechanical coupling factors were obtained. They were shear wave velocity $v_s = 3.8 \times 10^5$ cm/sec, longitudinal one $v_l = 7.1 \times 10^5$ cm/sec, $k' = 0.68$ and $k_t = 0.44$, where k' is the function of k_{15} and k_{16} . In the frequency region above 100 Mc,

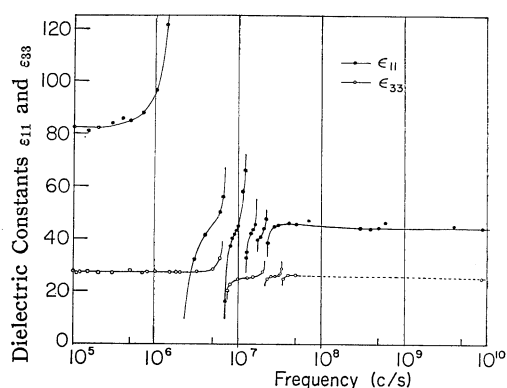


Fig. 1. Frequency dependence of dielectric constants, ϵ_{11} and ϵ_{33} .

the dielectric constants are again independent of frequency and it implies that they express the values ϵ^S corresponding to the clamped state. Loss tangent δ (ϵ''/ϵ') keeps a constant value over the whole frequency range except for acoustical resonance regions, where it exhibits nearly bell-shaped absorption characteristic to the resonance-type loss mechanism.

The difference between ϵ^T and ϵ^S is given by the following formula according to the thermodynamic theory

$$\epsilon^T_{ij} - \epsilon^S_{ij} = d_{ikl} d_{jmn} c^E_{klmn}, \quad (1)$$

where d means piezoelectric strain constant and c^E elastic stiffness under a constant electric field. By the use of measured values for $d^{(1)}$ and $c^{E(2)}$, it follows $\epsilon^{11T} - \epsilon^{11S} = 38.2$ and $\epsilon^{33T} - \epsilon^{33S} = 6.9$, while the values obtained from Fig. 1 are 38.5 and 2.5, respectively. Quite a good coincidence is found in ϵ_{11} , while a part of the discrepancy in ϵ_{33} may be attributed to the measured d_{33} value, since it is difficult to determine d_{33} accurately with the experimental procedures used in reference 1). The values of ϵ^S found in the present experiment coincides with those measured at far infrared region³⁾ and therefore, it is concluded that other dispersion mechanism than piezoelectric does not exist in the frequency range from D.C. to far infrared.

The dielectric constants were measured in the temperature range from 20 to 300°C at 9 Gc, and the results are shown in Fig. 2 with loss tangent

preparation of single crystals. Thanks are also due to Dr. T. Ikeda for valuable discussion.

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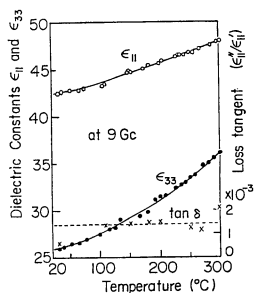


Fig. 2. Temperature dependence of dielectric constants and loss tangent δ .

δ . Dielectric constants rise with increasing temperature, while the loss tangent does not show any appreciable change in this temperature range.

Anisotropy of dielectric properties were measured at 9 Gc. Bars examined were cut out of (100) plate and the direction of the bar length was rotated around the x -axis of crystal. Both of ϵ' and ϵ'' well satisfy the angular variation expected from the transformation rule of a second-rank tensor, and ϵ_{11}'' attains 0.11 and ϵ_{33}'' 0.03.

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